

Course

Data Science Research Methods

Reproducible Research

Introduction

Topics

- Reproducible research
- Recap LaTeX
- Recap R
- Knitr: combining R and LaTeX

Reproducible Research

Reproducibility was mentioned in other course (?):

“Term coined by Jon Claerbout (a computer scientist); means that data sets and computer code be made available such that if we start from the data gathered by the scientist we can reproduce the same results, p-values, confidence intervals, tables and figures as those reported by the scientist”.

*“The results you present in your **assignments** should be reproducible by someone else (your future self in ten years, your grandmother, ...)”*

Statistical analysis and reports

Common problems with including statistical analysis results into reports include:

- time consuming to revise analysis (e.g. leave out outliers and re-analyse data) because it is not clear how results and graphs were produced
- difficult to reproduce results later because not clear which data sets and scripts were used

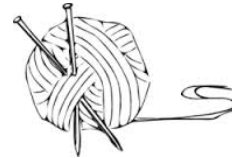
Solution: integrate text of report with code for statistical analysis.

Literate reporting and reproducible research

Concept of integrating report and computations was invented by computer scientist Knuth (“literate reporting”)

Introduced in statistics / R through work of Leisch and others (Sweave).

New, improved form by Hui through the R package knitr (graphics, caching,....)

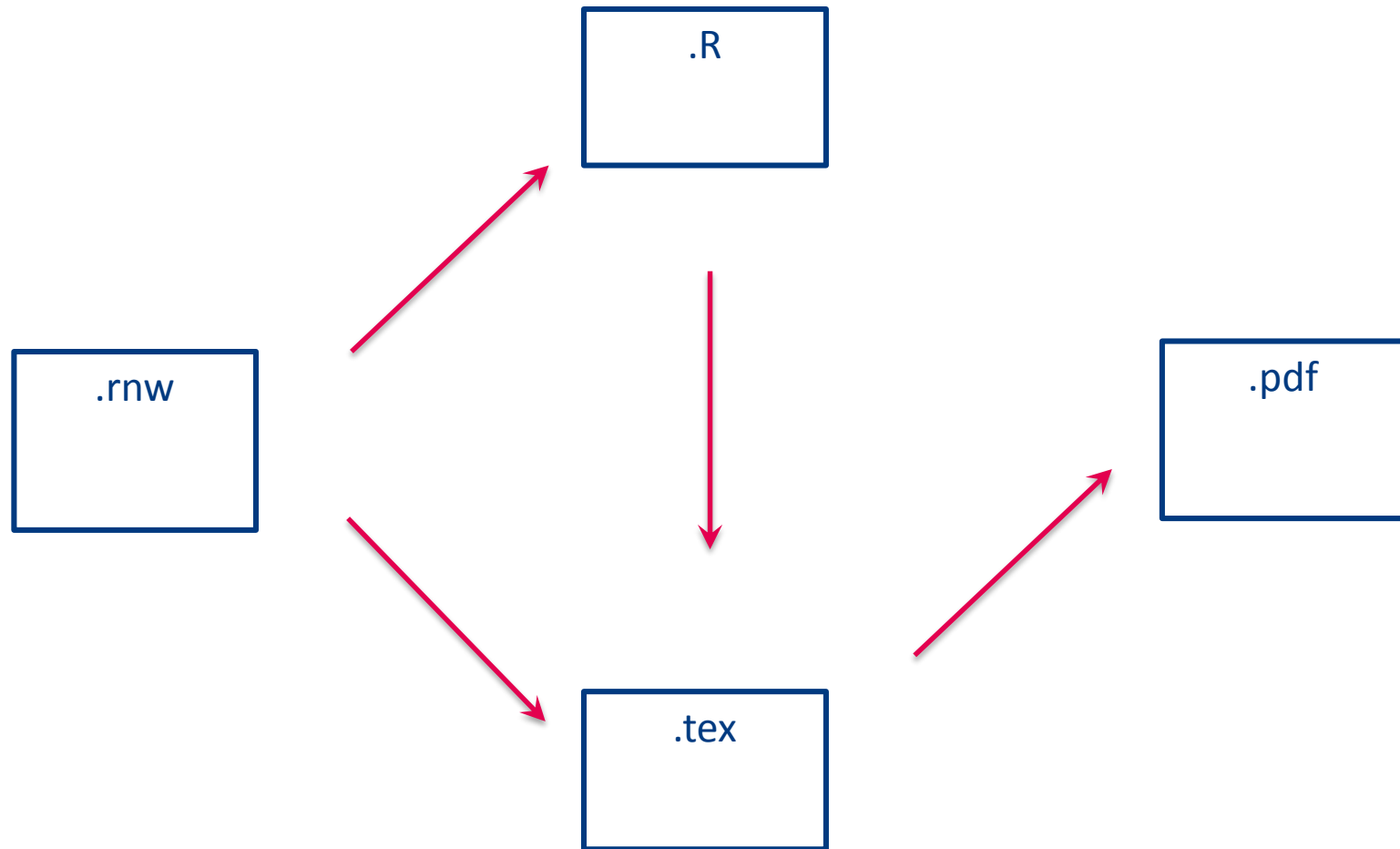


Convenient implementation in RStudio (“one-click button”) and Overleaf (but restriction w.r.t. packages)

This makes it possible to make research reproducible.

Also very useful for repeating, standard analyses and group work.

Knitr – schematic working



Recap LaTeX

- Skeleton
- Preamble
- Title page
- Sections and paragraphs
- Tables and figures
- Mathematics

LaTeX - skeleton

```
\documentclass[11pt]{article}

% comments on lines starting with %
% other document classes are report and book
% Specific classes like beamer for presentations
% they differ w.r.t. title pages and sectioning (chapters,
% sections,.....)

% here comes the preamble (see next slide)

\begin{document}
% here the main text should be written
\end{document}
```


LaTeX - preamble

% load extra packages for hyphenation, paper format, ...

% Highly recommended

```
\usepackage[margin=2cm,a4paper]{geometry}
```

% to change from default letter format (= “American A4”)

```
\usepackage[english]{babel} % for hyphenation
```

```
\usepackage{amsmath,amssymb} % extra math from AMS
```

```
\usepackage{graphicx} % nicer graphics
```

%Optional

```
\usepackage{fancyhdr} % fine-tuning of headers and footers
```

```
\usepackage{natbib} % for nice bibliographical references
```

```
\usepackage{lastpage} % page counts
```

LaTeX - title page

```
\title{A Student Guide to Writing Reports with \LaTeX}  
\author{Alessandro Di Bucchianico\Eindhoven University of  
Technology}  
\date{\today} % or \date{} to suppress date  
\maketitle
```

Optional:

```
\tableofcontents  
\listoffigures  
\listoftables
```

LaTeX – sections and paragraphs

Begin new sections/chapters/ etc. like this

```
\section{Introduction}\label{sec:intro}
```

Other options: part, chapter (only in books/reports),
subsection, subsection, ...

Paragraphs will be started by adding an empty line (this will
indent automatically as required by English writing rules!).

Do not use \\ !

Labels are useful since you can refer to the item using
`\ref{...}`. This will be updated automatically after recompiling.

LaTeX – tables and figures

Tables and figures are included as “floating environments”:

```
\begin{table}[htb]
\centering
\begin{tabular}{cc}
student & grade \\
... & ... \\
\end{tabular}
\caption{This is my table}\label{tab:demotable}
\end{table}
```

```
\begin{figure}[htb]
\centering
\includegraphics{filename} % if .pdf, remove extension
\caption{This is my figure}\label{fig:demofigure}
\end{figure}
```

LaTeX - mathematics

In-line equations with $\$ \dots \$$

Numbered equations with $\$ \$ \dots \$ \$$

NB No empty line after the last $\$ \$$

Useful commands:

$\backslash \text{frac}\{\}\{\}$ % fractions

There are numerous options to have multi-line equations,
constructs for matrices, ...

Recap R

- data structures
- data import
- assignment and selection of variables
- vectorised operations
- statistical tests
- objects (density, hist, lm)
- plots

R – data structures

vector (one-dimensional data structure, `c` = concatenate):

- `c(1,2,3,7,"hello")`
- `c("Joost","Jeroen","Nard","Eric","Hoy-Ying")`

matrix (all columns of same type):

- `matrix(c(1,2,3,4,5,6),ncol=2,byrow=T)`

data frame (all columns of same length):

- `data.frame(name=c("Joost","Nard","Eric","Hoy-Ying"),
phone=c(3455,3175,3528,3551),gender=c("M","M","M","F"))`

list (most general data type, no restrictions):

- `list(name= c("Joost","Nard"),pc=c(5503,5521,5534))`

Use **str** to get an overview of a data structure

R – data import

scan for reading in a sequence of numbers

read.table for reading data sets with several columns

Useful option: header = T

Import from Excel is possible, but it is safer to use .csv format

.csv = **comma separated values**. It is preferred to use ; as separator instead of , because English uses decimal points (e.g., 2.3) while other languages use commas: 2,3

read.csv or **read.csv2** for importing .csv files

R – assignment and selection: vectors

`a <- 2 + sqrt(5)`

`x[3]` # 3rd element of x

`x[-3]` # all but 3rd element of x

`x[3:5]` # elements 3 to 5

`x[x<= 3.2]` # all elements less than or equal to 3.2

`x[x == "M"]` # all elements equal to character M

`head(x)` # the first six elements of x

R - selection data frames

age	length
22	136
13	145
24	185

mydata[,1]

mydata[1,2]

mydata[3,]

Columns can also be selected as `mydata$age` or `mydata["age"]`. Also useful: `head(mydata)`, `tail(mydata)` (first/last 6 rows), `dim(mydata)`, `names(mydata)`, `attach(mydata)` (allows direct calling of column names)

R – vectorised operations

Elementwise operations on vectors are a powerful feature of R. They can make your code short and easy to read.

Examples:

$(1:10)^2$

Short way to compute sample variance (also available through the standard functions **var** and **sd**)

$\text{sum}((x - \text{mean}(x))^2) / (\text{length}(x) - 1)$

R – statistical tests

t.test(x, alternative = "two.sided", mu = 5, conf.level = 0.95)

t.test(x, y , alternative = "less", mu = 0, paired = FALSE,
var.equal = FALSE, conf.level = 0.95)

binom.test(successes,samplesize,p=0.3,...)

Normality testing

qqnorm followed by **qqline** (normal probability plot)

plot(density(..)) (density estimator to detect shape deviations)

shapiro.test (for the Shapiro-Wilk test)

- \$p.value
- \$statistic

R – objects

Many procedures in R produce an object, e.g.

hist

density

t.test

Details of the object can be inspected through **str**.

The idea is that you manipulate or extract information from objects by using the **\$** operator or by applying functions (called **methods** in computer science jargon), e.g.

plot

summary

R - plots

plot - basic function for scatterplots, but also kernel density estimators and histograms

axis – allows to fine tune an axis (requires to suppress an axis in plot by using `xaxt="n"` or `yaxt="n"`)

abline – draw horizontal or vertical line

legend – add legend

Functions may be drawn using `curve`:

curve(x^2-3 , from=0,to=2)

Knitr

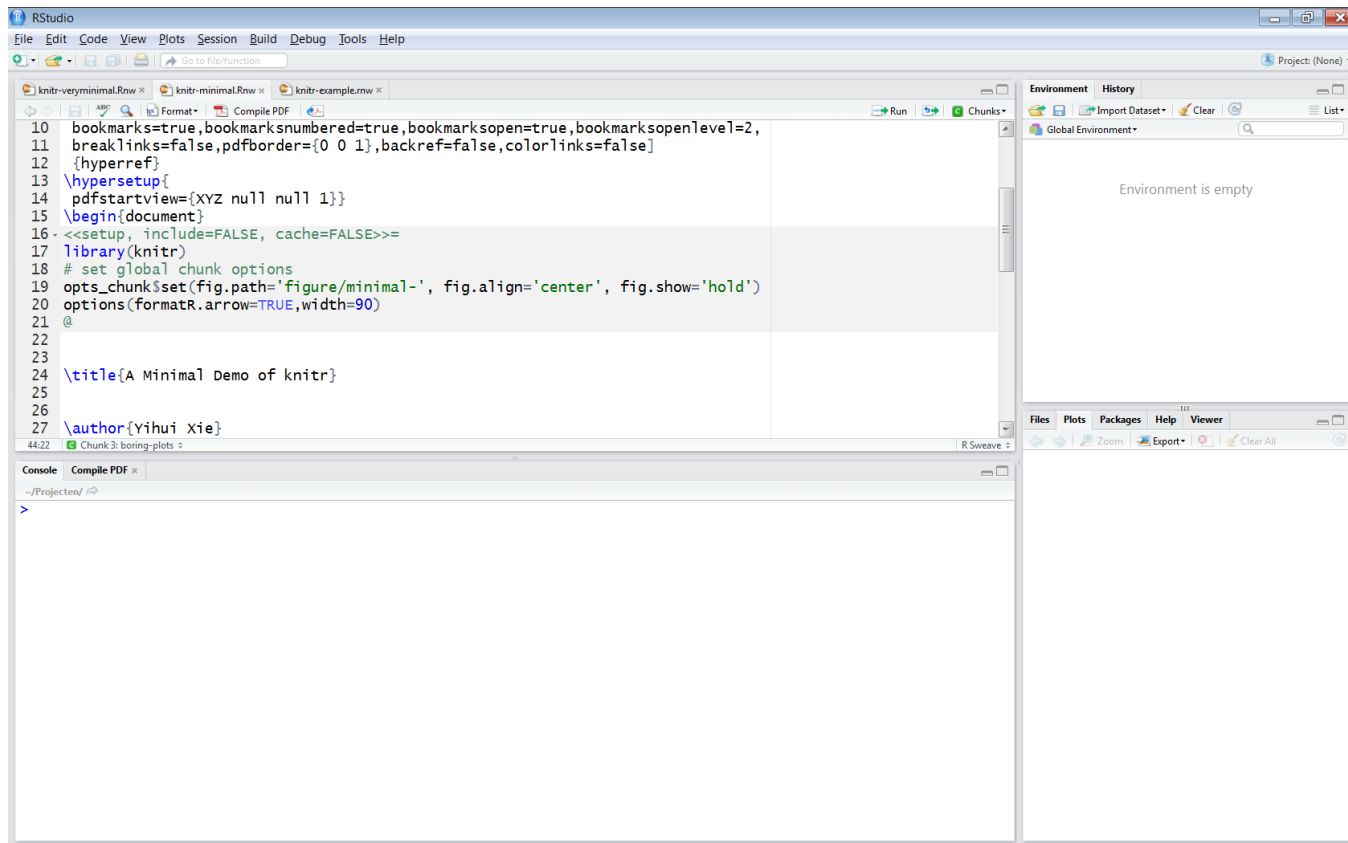
- basic implementation
- chunks
- adaptations to Rstudio
- Overleaf

Basic implementation

- one file which includes both the text (through LaTeX) and the statistical analysis (R scripts in the form of “chunks”)
- for historical reasons the extension of the file is .rnw (stands for R NoWeb)
- requires knitr package
- .tex is automatically created, use `purl("file.rnw")` to get an .R file with all the R commands

More information: <http://yihui.name/knitr/>

Example of .rnw file in RStudio



The screenshot shows the RStudio interface with a .rnw file open. The code in the editor is as follows:

```
10 bookmarks=true,bookmarksnumbered=true,bookmarksopen=true,bookmarksopenlevel=2,
11 breaklinks=false,pdfborder={0 0 1},backref=false,colorlinks=false]
12 {hyperref}
13 \hypersetup{
14 pdfstartview={XYZ null null 1}}
15 \begin{document}
16 <<setup, include=FALSE, cache=FALSE>>=
17 library(knitr)
18 # set global chunk options
19 opts_chunk$set(fig.path='figure/minimal-', fig.align='center', fig.show='hold')
20 options(formatR.arrow=TRUE,width=90)
21 @
22
23
24 \title{A Minimal Demo of knitr}
25
26
27 \author{Yihui Xie}
```

The RStudio interface includes a menu bar at the top, a toolbar with icons for file operations, a source editor on the left, a console at the bottom, and a right-hand pane with tabs for Environment, History, Files, Plots, Packages, Help, and Viewer. The Environment tab is active, showing "Global Environment" and "Environment is empty". The console shows the current directory as "~/Projecten/".

Chunks and \Sexpr

R code in “chunks” (again, form for historical reasons):

```
<<nameofchunk, options>>=
```

```
your R code
```

```
@
```

It is important to give clear, distinct names to chunks

- makes debugging easy (to locate errors)
- essential to get labels to plots

RStudio has an automated way to show chunk options.

Computations in the text are possible through the \Sexpr command

Location of data files and figures

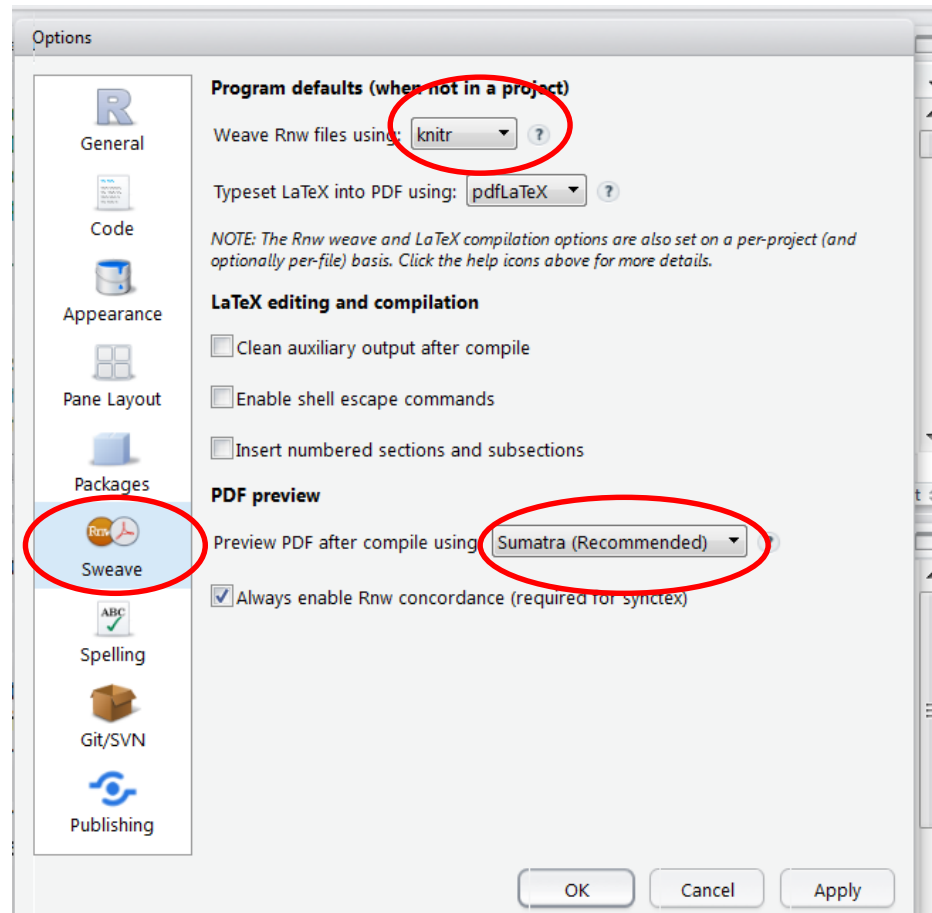
It is good custom to store figures in a separate folder.

Knitr will automatically set the working directory of R to the file location. Do not use `setwd` as this will be overwritten. If you use relative paths in R chunks, then you do not need to set paths.

For simple documents, put your data file in the same folder as your `.rnw` document.

Putting figures in a separate folder can be set as a global option (see previous slide). Note the use of a relative path.

Knitr - adapting RStudio (Windows)



Knitr - Adobe and Sumatra

Adobe Acrobat sometimes causes problems since it does not allow other software to make modifications to a pdf file which is open.

Sumatra is a free pdf viewer (for Windows) that does not have this problem.

Knitr - installing required extra packages

It is recommended to update R to the latest version before installing new packages

The screenshot shows the RStudio interface with a Knitr document open. The code editor displays R code for generating a summary table. A dialog box titled "Installing Packages" is open, showing the repository (CRAN) and the library path (D:/Users/adibucchi/Documents/R/win-library/3.2 [Default]). The console shows the output of the Knitr process, including the creation of a PDF file.

Code in Editor:

```
68 @
69
70 <<summarytablewithoutlier,results='asis',echo=FALSE>>=
71 # turn data into dataframe
72 statsnames <- c("mean","variance","standard deviation","median")
73 statsvaluesclean <- formatC(c(mean(dataclean), var(dataclean),sd(dataclean),median(dataclean)),format
74 ="f",digits=2)
75 datatableclean <- rbind(statsnames,statsvaluesclean)
76 print(xtable(datatableclean,caption="Summary statistics without outlier in table form",label="tab
77 :withoutoutlier"),hline.after=c(1),include.colnames=TRUE)
78
79 \noindent We check normality of the data by making
80 \begin{figure}[ht]
81 <<normalityplots,echo=FALSE,fig.width=6>>=
82 par(mfrow=c(1,2))
83
```

Console Output:

```
Chunk 7: summarytablewithoutlier
ecno: logi FALSE

|.....| 95%
label: summarytableCelsius (with options)
List of 2
 $ results: chr "asis"
 $ echo   : logi FALSE

|.....| 100%
ordinary text without R code

output file: HW1Exercisel.tex
[1] "HW1Exercisel.tex"
Running pdflatex.exe on HW1Exercisel.tex...completed
Created PDF: ~/Onderwijs/2WS30/Homework/HW1Exercisel.pdf
```

Environment Panel:

Global Environment
Data
a 35 obs. of 1 variable
dataclean chr [1:2, 1:4] "mean..."
mydata 35 obs. of 1 variable
Values
data num [1:35] 84 49 61 4...
dataclean num [1:35] 84 49 61 4...
statsnames chr [1:4] "mean" "var..."
statsvaluesclean chr [1:4] "66.86" "11..."

User Library:

Name	Description	Version
abind	Combine Multidimensional Arrays	1.4-3
acepack	ace() and avas() for selecting regression transformations	1.3-
aplpack	Another Plot Package: stem, leaf, bagplot, faces, spin3R, plotsummary, plotthulls, and some slider functions	1.3.0
arm	Data Analysis Using Regression and Multilevel/Hierarchical Models	1.8-6
biglm	bounded memory linear and generalized linear models	0.9-1
bitops	Bitwise Operations	1.0-6
brew	Templating Framework for Report Generation	1.0-6
BsMD	Bayes Screening and Model Discrimination	2013.07
calibrate	Calibration of Scatterplot and Biplot Axes	1.7-2
car	Companion to Applied Regression	2.0-25
caTools	Tools: moving window statistics, GIF, Base64, ROC AUC, etc.	1.17.1
chron	Chronological objects which	2.3-45

Overleaf

Overleaf is an on-line LaTeX editor.

It has the option to work with knitr.

Files need to have the extension .Rtex

Please note that you cannot install libraries but many widely used packages have been installed.



Summary

- Reproducible research
- LaTeX:
 - preamble
 - sections and referencing by labels
 - floating environments
- R
 - data structures
 - import data
 - objects
 - vectorized operations
- Knitr
 - chunks